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Higher Education Process Flows — CAUDIT HERM Reference Model

Generated on 2026-07-06 | Domain: Higher Education | Framework: CAUDIT Higher Education Reference Models (HERM)

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Test Data

The following test data is used consistently across all sequence diagrams:

Entity	Field	Value
Student	id	STU-2026-001
Student	name	John Doe
Student	email	john.doe@university.edu
Student	program	B.Sc. Computer Science
Student	semester	Fall 2026
Student	gpa	3.75
Prospect	id	PRO-2026-042
Prospect	name	Jane Smith
Prospect	email	jane.smith@prospect.com
Prospect	program	M.Sc. Data Science
Prospect	source	Open Day
Course	id	CS501
Course	title	Artificial Intelligence & ML
Course	credits	4
Course	instructor	Dr. Sarah Smith
Course	capacity	120
Course	enrolled	98
Faculty	id	FAC-101
Faculty	name	Dr. Sarah Smith
Faculty	dept	Computer Science
Faculty	email	sarah.smith@university.edu
Faculty	hire_date	2024-08-15
Faculty	salary	95000
Finance	fee_amount	12500.0

Entity	Field	Value
Finance	scholarship	2500.0
Finance	grant_amount	5000.0
Finance	invoice_no	INV-2026-8901
Finance	payment_ref	PAY-7X3K2
Research	grant_id	GR-2026-045
Research	title	AI-Driven Personalized Learning
Research	amount	250000.0
Research	sponsor	NSF
Research	pi	Dr. Sarah Smith
Research	status	Active
Alumni	id	ALUM-2019-088
Alumni	name	Emily Johnson
Alumni	grad_year	2019
Alumni	degree	B.Sc. Computer Science
Alumni	email	emily.j@corp.com

Student Administration

End-to-end management of the student lifecycle from prospect engagement through graduation.

1.1 Student Onboarding (Prospect → Enrollment)

Description: A prospective student expresses interest, submits an application, is admitted, accepts the offer, enrolls, and registers for courses.

Actors: Prospect/Student, Admissions Officer

Systems: CRM, SIS, LMS, Finance, Housing

Sequence Diagram

```
sequenceDiagram
    participant P as Prospect/Student
    participant CRM as CRM<br/>(Salesforce)
    participant SIS as SIS<br/>(Ellucian Banner)
```

participant LMS as LMS
(Canvas)
participant FIN as Finance
(Workday)
participant HOUS as Housing
P->>CRM: Submit inquiry
name=Jane Smith
program=MSc Data Science
CRM->>CRM: Create prospect record
ID: PRO-2026-042
P->>CRM: Submit application
transcripts, essays
CRM->>SIS: Create applicant
createApplicant(PRO-2026-042)
SIS-->>CRM: ApplicantID: APP-2026-088
Admissions Officer->>SIS: Review & admit
decision=Accepted
SIS->>CRM: Update status
status=Admitted
CRM->>P: Offer letter
program=MSc Data Science
P->>CRM: Accept offer
CRM->>SIS: Confirm enrollment
enrollStudent(APP-2026-088)
SIS->>FIN: Create fee account
createFeeAccount(STU-2026-001, \$12,500)
SIS->>LMS: Create user & course access
createCanvasUser(STU-2026-001)
SIS->>HOUS: Request accommodation
housingRequest(STU-2026-001)
HOUS-->>SIS: Room assigned
Hall: Adams, Room 204
SIS->>P: Welcome email with credentials

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Student Onboarding	Prospect	CreateProspect	CRM (Salesforce Education Cloud)	SIS (Ellucian Banner)	REST API (OAuth2)	REST API (API Key)	API-led (Real-time)	Medium
Student Onboarding	Application	SubmitApplication	CRM (Salesforce Education Cloud)	SIS (Ellucian Banner)	REST API (OAuth2)	REST API (API Key)	API-led (Real-time)	Medium
Student Onboarding	Student	EnrollStudent	SIS (Ellucian Banner)	LMS (Canvas)	Banner API (Basic Auth)	Canvas API (OAuth2)	Event-driven	Simple
Student Onboarding	Fee Account	CreateFeeAccount	SIS (Ellucian Banner)	Finance ERP (Workday Financials)	Banner Workflow (SOAP)	Workday API (WS-Security)	API-led (Real-time)	Medium

1.2 Enrollment & Registration

Description: Existing students register for courses each semester. System validates prerequisites, checks seat availability, processes fees, and confirms enrollment.

Actors: Student, Registrar
Systems: Student Portal, SIS, LMS, Finance

Sequence Diagram

```
sequenceDiagram
    participant STU as Student
    participant PORT as Student Portal
    participant SIS as SIS<br/>(PeopleSoft)
    participant FIN as Finance<br/>(Oracle EBS)
    participant LMS as LMS<br/>(Blackboard)
    STU->>PORT: Login & select courses<br/>CS501, MATH302, ENG201
    PORT->>SIS: Check prerequisites<br/>checkPrereqs(STU-2026-001, [CS501,...])
    SIS-->>PORT: Prerequisites satisfied
    PORT->>SIS: Check seat availability<br/>checkSeats([CS501,...])
    SIS-->>PORT: Seats available
    STU->>PORT: Confirm registration
    PORT->>SIS: RegisterStudent(STU-2026-001, [CS501,...])
    SIS->>SIS: Calculate tuition<br/>total=$12,500.00
    SIS->>FIN: Create invoice<br/>createInvoice(STU-2026-001, $12,500)
    FIN-->>SIS: InvoiceID: INV-2026-8901
    PORT->>FIN: Process payment<br/>creditCard(ref=PAY-7X3K2, amt=$12,500)
    FIN-->>PORT: Payment confirmed
    FIN->>SIS: Payment confirmation
    SIS->>LMS: Enroll in courses<br/>createEnrollments(STU-2026-001, [CS501,...])
    LMS-->>SIS: Enrollments confirmed
    SIS->>PORT: Registration complete<br/>schedule, invoice, receipts
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Enrollment & Registration	Course Enrollment	RegisterCourse	Student Portal	SIS (PeopleSoft Campus Solutions)	Portal REST API	PeopleSoft REST API (JWT)	API-led (Real-time)	Simple
Enrollment & Registration	Invoice	CreateInvoice	SIS (PeopleSoft Campus Solutions)	Finance ERP (Oracle EBS)	PeopleSoft Integration Broker	Oracle EBS API (SOAP)	API-led (Real-time)	Medium
Enrollment & Registration	Payment	ProcessPayment	Payment Gateway (TouchNet)	Finance ERP (Oracle EBS)	Payment Gateway API	Oracle EBS REST API	Event-driven	High
Enrollment & Registration	Enrollment	SyncEnrollmentToLMS	SIS (PeopleSoft Campus Solutions)	LMS (Blackboard Learn)	PeopleSoft REST API (JWT)	Blackboard REST API (OAuth2)	Batch (Scheduled)	Simple

Student Administration

1.3 Academic Records Management

Description: Faculty submit grades. SIS calculates GPA, updates transcripts, notifies students. Transcript requests fulfilled digitally.

Actors: Faculty, Student, Registrar

Systems: LMS, SIS, Student Portal, Data Warehouse

Sequence Diagram

```
sequenceDiagram
    participant FAC as Faculty
    participant LMS as LMS<br/>(Canvas)
    participant SIS as SIS<br/>(Ellucian Banner)
    participant PORT as Student Portal
    participant DW as Data Warehouse<br/>(Snowflake)
    FAC->>LMS: Submit final grades<br/>CS501: A, MATH302: B+, ...
    LMS->>SIS: Push grades<br/>gradePush(courseID=CS501, term=Fall2026)
    SIS->>SIS: Calculate GPA<br/>student=STU-2026-001, GPA=3.75
    SIS->>SIS: Update transcript
    SIS->>PORT: Notify grades available
    PORT->>SIS: View transcript
    SIS-->>PORT: Display transcript
    Student->>PORT: Request official transcript
    PORT->>SIS: generateTranscript(STU-2026-001, official=true)
    SIS-->>PORT: Digital transcript (PDF)
    SIS->>DW: Incremental load grades<br/>table: fact_grades
    DW->>DW: Update student_rollup
    LMS->>DW: Load course activity<br/>table: fact_course_activity
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Academic Records	Grade	GradePush	LMS (Canvas)	SIS (Ellucian Banner)	Canvas API (OAuth2)	Banner API (API Key)	Batch (Scheduled)	Simple
Academic Records	Transcript	GenerateTranscript	SIS (Ellucian Banner)	Student Portal	Banner REST API	Portal REST API	API-led (Real-time)	Simple
Academic Records	Grade	GradeToDW	SIS (Ellucian Banner)	Data Warehouse (Snowflake)	Banner ODBC / CDC	Snowflake JDBC	Batch (ETL)	Medium

1.4 Graduation & Certification

Description: Students apply to graduate. System audits degree requirements, clears financial holds, confers degree, issues digital diploma.

Actors: Student, Registrar

Systems: Student Portal, SIS, Finance, Alumni CRM

Sequence Diagram

```
sequenceDiagram
    participant STU as Student
    participant PORT as Student Portal
    participant SIS as SIS<br/>(Ellucian Banner)
    participant FIN as Finance<br/>(Workday)
    participant CRM as Alumni CRM
    STU->>PORT: Apply for graduation<br/>semester=Spring 2026, degree=B.Sc. CS
    PORT->>SIS: submitGraduationApp(STU-2026-001)
    SIS->>SIS: Degree audit<br/>check 120 credits, GPA>=2.0
    SIS-->>PORT: Requirements satisfied
    SIS->>FIN: Check financial clearance<br/>checkHolds(STU-2026-001)
    FIN-->>SIS: No holds
    Registrar->>SIS: Approve graduation
    SIS->>SIS: Confer degree & generate diploma
    SIS->>PORT: Degree conferred, download diploma
    SIS->>CRM: Create alumni record<br/>createAlumni(STU-2026-001, B.Sc. CS, 2026)
    CRM->>CRM: Welcome kit, alumni benefits
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Graduation & Certification	Graduation Application	SubmitGradApp	Student Portal	SIS (Ellucian Banner)	Portal REST API	Banner REST API (API Key)	API-led (Real-time)	Simple
Graduation & Certification	Financial Clearance	CheckFinancialHolds	SIS (Ellucian Banner)	Finance ERP (Workday Financials)	Banner Workflow (SOAP)	Workday API (WS-Security)	API-led (Real-time)	Medium
Graduation & Certification	Alumni Record	CreateAlumniRecord	SIS (Ellucian Banner)	Alumni CRM (Blackbaud)	Banner REST API	Blackbaud API (OAuth2)	Event-driven	Simple

Learning & Teaching

Design, develop, deliver, and assess academic programs and courses. Spans curriculum planning, timetable scheduling, delivery, and assessment.

2.1 Curriculum Management

Description: Academic committees design new programs/courses. Approved curricula published to SIS catalog and LMS.

Actors: Curriculum Committee, Dean, Registrar

Systems: Curriculum Mgmt System, SIS, Student Portal, LMS

Sequence Diagram

```
sequenceDiagram
    participant CC as Curriculum Committee
    participant CMS as Curriculum Mgmt<br/>(CourseLeaf)
    participant SIS as SIS<br/>(PeopleSoft)
    participant PORT as Student Portal
    participant LMS as LMS<br/>(Canvas)
    CC->>CMS: Propose new course<br/>CS502: Deep Learning, 4 credits
    CMS->>CMS: Workflow routing<br/>department to school to senate
    Dean->>CMS: Approve proposal
    CMS->>SIS: Create course catalog entry<br/>createCourse(CS502, Deep Learning, 4cr)
    SIS-->>CMS: CourseID: CS502
    CMS->>SIS: Define program structure<br/>B.Sc. CS requires CS502
    SIS->>PORT: Publish updated catalog
    SIS->>LMS: Create course shell<br/>createCourseShell(CS502, Fall2026)
    CC->>CMS: Update syllabus<br/>learning outcomes, assessments
    CMS->>LMS: Sync syllabus<br/>syllabusID: SYL-2026-014
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Curriculum Management	Course	CreateCourse	Curriculum Mgmt (CourseLeaf)	SIS (PeopleSoft Campus Solutions)	CourseLeaf CIM API	PeopleSoft REST API (JWT)	API-led (Real-time)	Medium
Curriculum Management	Course Shell	CreateCourseShell	SIS (PeopleSoft Campus Solutions)	LMS (Canvas)	PeopleSoft REST API (JWT)	Canvas API (OAuth2)	Event-driven	Simple
Curriculum Management	Syllabus	SyncSyllabus	Curriculum Mgmt (CourseLeaf)	LMS (Canvas)	CourseLeaf REST API	Canvas API (OAuth2)	Batch (Scheduled)	Simple

2.2 Course Delivery & Assessment

Description: Courses delivered via LMS. Students submit assignments, take quizzes. Faculty grade and final grades push to SIS.

Actors: Student, Faculty

Systems: LMS, Assessment Tool, SIS

Sequence Diagram

```
sequenceDiagram
    participant STU as Student
    participant LMS as LMS<br/>(Blackboard)
    participant ASSESS as Assessment Tool<br/>(Proctorio)
    participant SIS as SIS<br/>(Ellucian Banner)
    LMS->>LMS: Publish course materials<br/>CS501 Week 1-14 modules
    STU->>LMS: Access lecture videos and readings
    STU->>LMS: Submit assignment<br/>A1: Neural Networks Essay
    LMS->>ASSESS: Send for AI similarity check
    ASSESS-->>LMS: Similarity score: 12 percent
    FAC->>LMS: Grade assignment<br/>score: 88/100
    LMS->>LMS: Update gradebook
    STU->>LMS: Take online quiz<br/>Midterm: 42/50
    LMS->>ASSESS: Proctoring validation
    ASSESS-->>LMS: Proctoring result: Pass
    FAC->>LMS: Finalize course grades
    LMS->>SIS: Push final grades<br/>gradePush(CS501, Fall2026)
    SIS-->>LMS: Grades accepted
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Course Delivery & Assessment	Assignment Submission	SubmitAssignment	LMS (Blackboard Learn)	Assessment Tool (Proctorio)	Blackboard REST API (OAuth2)	Proctorio API (API Key)	API-led (Real-time)	Medium
Course Delivery & Assessment	Final Grade	GradePushToSIS	LMS (Blackboard Learn)	SIS (Ellucian Banner)	Blackboard REST API (OAuth2)	Banner API (API Key)	Batch (Scheduled)	Simple

2.3 Timetable & Scheduling

Description: Schedulers create master timetable, assign rooms, check conflicts, publish schedules.

Actors: Scheduler, Faculty, Student

Systems: Scheduling Engine, SIS, Room Booking, Portal

Sequence Diagram

```
sequenceDiagram
    participant SCH as Scheduler
    participant ENG as Scheduling Engine<br/>(Ad Astra)
    participant SIS as SIS<br/>(PeopleSoft)
    participant ROOM as Room Booking<br/>(EMS)
    participant PORT as Student Portal
    SCH->>ENG: Create timetable<br/>semester=Fall2026, term=16-week
    ENG->>SIS: Load course demand<br/>getCourseDemand(Fall2026)
    SIS-->>ENG: 1,250 course requests
    ENG->>ROOM: Check room availability<br/>getAvailableRooms(time_slots)
    ROOM-->>ENG: Available rooms list
    ENG->>ENG: Optimize schedule<br/>minimize conflicts
    ENG-->>SCH: Proposed timetable
    SCH->>ENG: Publish timetable
    ENG->>SIS: Sync scheduled sections<br/>syncSections(Fall2026)
    SIS->>PORT: Publish student schedule
    FAC->>ENG: View teaching assignment<br/>CS501: Mon/Wed 10-11:30, Hall B
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Timetable & Scheduling	Course Demand	LoadCourseDemand	SIS (PeopleSoft Campus Solutions)	Scheduling Engine (Ad Astra)	PeopleSoft REST API (JWT)	Ad Astra API (API Key)	Batch (Scheduled)	Simple
Timetable & Scheduling	Room Availability	CheckRoomAvailability	Scheduling Engine (Ad Astra)	Room Booking (EMS)	Ad Astra REST API	EMS Web API (Basic Auth)	API-led (Real-time)	Medium
Timetable & Scheduling	Scheduled Section	SyncSections	Scheduling Engine (Ad Astra)	SIS (PeopleSoft Campus Solutions)	Ad Astra REST API	PeopleSoft REST API (JWT)	Batch (Scheduled)	Simple

HR & Faculty Management

Full lifecycle of faculty and staff: recruitment, onboarding, payroll, performance management, professional development.

3.1 Faculty Recruitment

Description: Departments request new positions. HR posts job, screens applicants, coordinates interviews, manages offer/acceptance.

Actors: Hiring Manager, HR, Candidate

Systems: HRIS, ATS, Finance, SIS

Sequence Diagram

```
sequenceDiagram
    participant HM as Hiring Manager<br/>(Dean)
    participant HR as HR<br/>(Workday HCM)
    participant ATS as ATS<br/>(Interfolio)
    participant FIN as Finance<br/>(Workday Financials)
    participant SIS as SIS
    HM->>HR: Submit faculty requisition<br/>position=Prof. CS, rank=Assistant
    HR->>FIN: Budget check<br/>checkBudget(CS, salary=$95,000)
    FIN-->>HR: Budget approved
    HR->>HR: Create position<br/>positionID: POS-2026-042
    HR->>ATS: Post job<br/>jobID: JOB-2026-088
    Candidate->>ATS: Submit application<br/>name=Dr. Sarah Smith
    ATS->>ATS: Initial screening
    HM->>ATS: Review shortlist
    HM->>HR: Interview scorecard<br/>score: 4.8/5.0
    HR->>HR: Generate offer letter<br/>salary=$95,000, start=Aug 15
    Candidate->>HR: Accept offer
    HR->>HR: Onboard new hire<br/>status=Active
    HR->>SIS: Create faculty profile<br/>ID: FAC-101
    SIS->>HR: FacultyID confirmed
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Faculty Recruitment	Position Requisition	SubmitRequisition	HRIS (Workday HCM)	Finance ERP (Workday Financials)	Workday API (WS-Security)	Workday API (WS-Security)	API-led (Real-time)	Simple
Faculty Recruitment	Job Posting	PostJob	HRIS (Workday HCM)	ATS (Interfolio)	Workday Recruiting API	Interfolio API (OAuth2)	API-led (Real-time)	Medium
Faculty Recruitment	Faculty Profile	CreateFacultyProfile	HRIS (Workday HCM)	SIS (Ellucian Banner)	Workday API (WS-Security)	Banner API (API Key)	Event-driven	Simple

3.2 Payroll & Benefits

Description: Monthly payroll: time/attendance to payroll, calculations, payslips, GL journal entries.

Actors: Faculty, HR, Finance

Systems: HRIS, Payroll, Finance, Time Tracking

Sequence Diagram

```
sequenceDiagram
    participant FAC as Faculty
    participant TIME as Time Tracking
    participant HR as HR<br/>(Workday HCM)
    participant PAY as Payroll System<br/>(ADP)
    participant FIN as Finance<br/>(Oracle EBS)
    FAC->>TIME: Submit timesheet<br/>week=Oct 5-11, hours=40
    TIME->>HR: Approve timesheet
    HR->>HR: Process payroll<br/>month=October 2026
    HR->>PAY: Submit payroll data<br/>employeeData(FAC-101, salary=$95k)
    PAY->>PAY: Calculate payroll<br/>gross=$7,916.67, deductions...
    PAY-->>HR: Payroll summary
    HR->>FAC: Payslip notification
    PAY->>FIN: Journal entry<br/>GL: salary expense $7,916.67
    FIN-->>PAY: Journal posted
    PAY->>FAC: Direct deposit<br/>net amount=$5,850.00
```

Integration Details

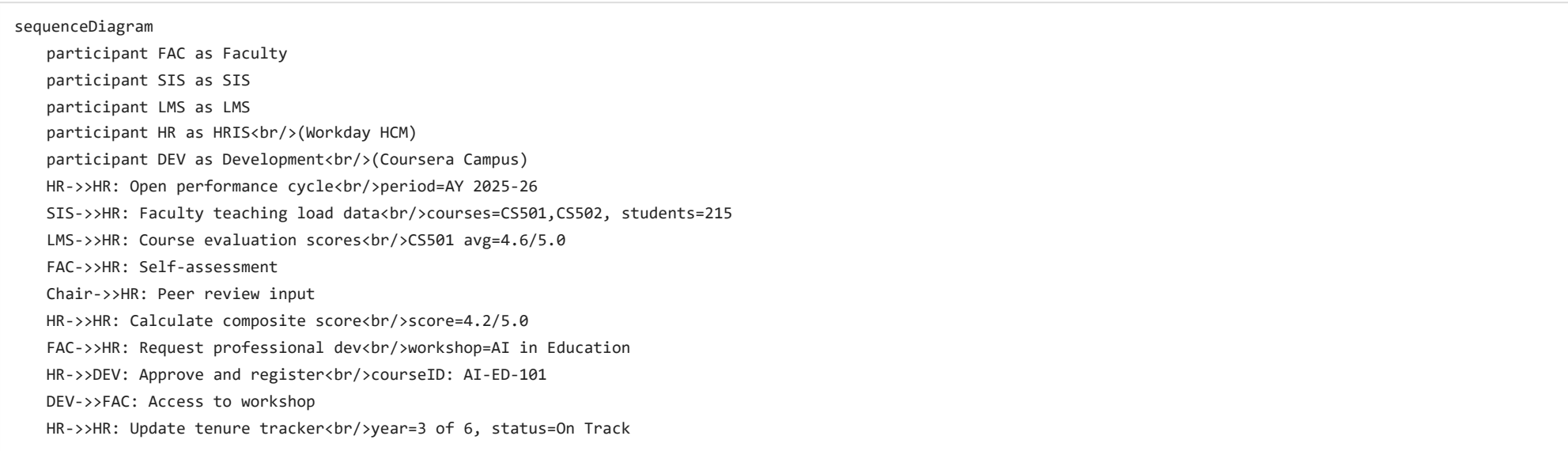
Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Payroll & Benefits	Timesheet	SubmitTimesheet	Time Tracking System	HRIS (Workday HCM)	Time API (REST)	Workday API (WS-Security)	API-led (Real-time)	Simple
Payroll & Benefits	Payroll Data	SubmitPayroll	HRIS (Workday HCM)	Payroll System (ADP)	Workday Payroll Interface	ADP API (SFTP)	Batch (Scheduled)	High
Payroll & Benefits	Journal Entry	PostGLJournal	Payroll System (ADP)	Finance ERP (Oracle EBS)	ADP GL Interface (SFTP)	Oracle EBS REST API (SOAP)	Batch (Scheduled)	Medium

3.3 Faculty Performance & Development

Description: Performance reviews, professional development, tenure tracking with inputs from SIS and LMS.

Actors: Faculty, Department Chair, HR
Systems: HRIS, LMS, SIS, Development Platform

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Faculty Performance & Development	Teaching Load	GetTeachingLoad	SIS (Ellucian Banner)	HRIS (Workday HCM)	Banner REST API	Workday API (WS-Security)	Batch (Scheduled)	Simple
Faculty Performance & Development	Course Evaluation	SyncEvalScores	LMS (Canvas)	HRIS (Workday HCM)	Canvas API (OAuth2)	Workday API (WS-Security)	Batch (Scheduled)	Medium
Faculty Performance & Development	Development Request	ApproveTraining	HRIS (Workday HCM)	Development Platform (Coursera Campus)	Workday API (WS-Security)	Coursera API (OAuth2)	API-led (Real-time)	Simple

Finance & Resource Management

Manage financial resources: student fees, financial aid, procurement, budgeting, general ledger.

4.1 Fee Assessment & Billing

Description: Student fees assessed per term. Invoices generated, payments processed, holds cleared.

Actors: Student, Finance, Bursar

Systems: SIS, Finance ERP, Payment Gateway, Student Portal

Sequence Diagram

```
sequenceDiagram
    participant SIS as SIS<br/>(PeopleSoft)
    participant FIN as Finance ERP<br/>(Workday Financials)
    participant PG as Payment Gateway<br/>(TouchNet)
    participant PORT as Student Portal
    SIS->>SIS: Calculate fees<br/>tuition=$10,000, fees=$2,500
    SIS->>FIN: Assess student fees<br/>assessFees(STU-2026-001, $12,500)
    FIN->>FIN: Generate invoice<br/>INV-2026-8901
    FIN->>PORT: Notify invoice available
    PORT-->>Student: View invoice
    Student->>PORT: Pay invoice<br/>credit card ending 4242
    PORT->>PG: processPayment($12,500)
    PG->>PG: Authorize and capture
    PG-->>PORT: Transaction approved<br/>ref: PAY-7X3K2
    PORT->>FIN: Payment notification
    FIN->>FIN: Update ledger<br/>AR decreased by $12,500
    FIN->>SIS: Clear fee hold
    PORT-->>Student: Receipt generated
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Fee Assessment & Billing	Student Fee	AssessFees	SIS (PeopleSoft Campus Solutions)	Finance ERP (Workday Financials)	PeopleSoft Integration Broker	Workday API (WS-Security)	API-led (Real-time)	Medium
Fee Assessment & Billing	Payment	ProcessPayment	Payment Gateway (TouchNet)	Finance ERP (Workday Financials)	TouchNet API	Workday API (WS-Security)	Event-driven	High
Fee Assessment & Billing	Fee Hold	ClearFeeHold	Finance ERP (Workday Financials)	SIS (PeopleSoft Campus Solutions)	Workday API (WS-Security)	PeopleSoft REST API (JWT)	Event-driven	Medium

4.2 Financial Aid Management

Description: Students apply for aid. System determines eligibility, packages aid, disburses funds.

Actors: Student, Financial Aid Officer, Finance

Systems: Financial Aid System, SIS, Finance ERP, Government Portal

Sequence Diagram

```
sequenceDiagram
    participant STU as Student
    participant FAN as Financial Aid(PowerFAIDS)
    participant GOV as Gov Portal(FAFSA/NSLDS)
    participant SIS as SIS(Ellucian Banner)
    participant FIN as Finance(Workday)
    STU->>FAN: Submit aid applicationFAFSA, institutional forms
    FAN->>GOV: Verify FAFSA datagetFederalData(SSN, tax info)
    GOV-->>FAN: EFC=$5,000, Pell eligible
    FAN->>FAN: Calculate needCOA=$30k - EFC=$5k = need $25k
    FAN->>SIS: Check enrollment statusgetEnrollment(STU-2026-001)
    SIS-->>FAN: Full-time, 15 credits
    FAN->>FAN: Package aidPell=$3,500, SEOG=$1,000, loan=$5,500
    FAN->>FAN: Award scholarshipMerit=$2,500
    FAN->>STU: Award lettertotal aid: $12,500
    STU->>FAN: Accept award
    FAN->>SIS: Post aid to accountpostAid(STU-2026-001, $12,500)
    FAN->>FIN: Disburse fundsdisburse(STU-2026-001, $12,500)
    FIN->>STU: Refund excessaid-fees = $0 (exact match)
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Financial Aid Management	FAFSA Data	VerifyFAFSA	Financial Aid System (PowerFAIDS)	Government Portal (FAFSA/NSLDS)	PowerFAIDS API	NSLDS Web Services (SOAP)	API-led (Real-time)	High
Financial Aid Management	Enrollment Status	CheckEnrollment	Financial Aid System (PowerFAIDS)	SIS (Ellucian Banner)	PowerFAIDS REST API	Banner API (API Key)	API-led (Real-time)	Simple
Financial Aid Management	Aid Disbursement	DisburseAid	Financial Aid System (PowerFAIDS)	Finance ERP (Workday Financials)	PowerFAIDS REST API	Workday API (WS-Security)	Event-driven	Medium

4.3 Procurement & Vendor Management

Description: Departments submit requisitions. Procurement processes POs, confirms receipt, matches invoices.

Actors: Department Admin, Procurement, Finance, Vendor

Systems: Procurement System, Finance ERP, Vendor Portal

Sequence Diagram

```
sequenceDiagram
    participant DEPT as Department Admin
    participant PROC as Procurement<br/>(Coupa)
    participant FIN as Finance<br/>(Oracle EBS)
    participant VEND as Vendor
    DEPT->>PROC: Submit requisition<br/>54-inch display x5, $1,200 ea
    PROC->>PROC: Approve requisition
    PROC->>VEND: Send PO<br/>PO-2026-301, total=$6,000
    VEND-->>PROC: PO accepted
    VEND->>PROC: Deliver goods<br/>5 displays delivered
    DEPT->>PROC: Confirm receipt
    PROC->>FIN: Create invoice payable<br/>invoice=INV-VEND-102, $6,000
    FIN->>FIN: 3-way match<br/>PO vs receipt vs invoice
    FIN->>FIN: Schedule payment<br/>net 30
    FIN->>VEND: Payment remittance<br/>wire transfer ref: W-8821
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Procurement & Vendor Management	Purchase Order	SendPO	Procurement System (Coupa)	Vendor Portal	Coupa API (OAuth2)	Vendor cXML / EDI	API-led (Real-time)	High
Procurement & Vendor Management	Invoice	SyncInvoice	Procurement System (Coupa)	Finance ERP (Oracle EBS)	Coupa API (OAuth2)	Oracle EBS REST API (SOAP)	Batch (Scheduled)	Medium

Research Management

Full research lifecycle: opportunity discovery, grant applications, award management, project execution, publication, compliance.

5.1 Grant Lifecycle Management

Description: Researchers identify funding, submit proposals, manage awarded grants, track expenditures, report to sponsors.

Actors: Researcher, Research Admin, Finance, Sponsor

Systems: Research Admin, Finance ERP, Compliance, Sponsor Portal

Sequence Diagram

```
sequenceDiagram
    participant RES as Researcher<br/>(Dr. Sarah Smith)
    participant RAD as Research Admin<br/>(InfoEd)
    participant FIN as Finance<br/>(Workday Financials)
    participant COMP as Compliance<br/>(IRB/IACUC)
    participant SPON as Sponsor<br/>(NSF)
    RES->>RAD: Find grant opportunity<br/>NSF AI Research, Due: 2026-03-15
    RES->>RAD: Submit proposal<br/>title=AI-Driven Personalized Learning
    RAD->>RAD: Internal review<br/>budget=$250k, IDC=25 percent
    RAD->>SPON: Submit proposal<br/>proposalID: NSF-2026-045
    SPON-->>RAD: Award notification<br/>grantID: GR-2026-045, $250k
    RAD->>FIN: Create award account<br/>createAwardAccount(GR-2026-045)
    RAD->>COMP: Register compliance<br/>human subjects, data privacy
    COMP-->>RAD: IRB approval: IRB-2026-102
    RES->>RAD: Submit expense report<br/>equipment=$45k, student stipend=$30k
    RAD->>FIN: Verify expense<br/>budget check: within limits
    FIN-->>RAD: Verified
    RAD->>SPON: Submit technical report<br/>Q1 progress, deliverables met
    SPON-->>RAD: Report accepted
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Grant Lifecycle Management	Proposal	SubmitProposal	Research Admin (InfoEd)	Sponsor Portal (NSF FastLane)	InfoEd API (OAuth2)	NSF FastLane API (Research.gov SOAP)	API-led (Real-time)	High
Grant Lifecycle Management	Award Account	CreateAwardAccount	Research Admin (InfoEd)	Finance ERP (Workday Financials)	InfoEd API (OAuth2)	Workday API (WS-Security)	Event-driven	Medium
Grant Lifecycle Management	Compliance	RegisterCompliance	Research Admin (InfoEd)	Compliance System (IRB Manager)	InfoEd REST API	IRB Manager API (API Key)	API-led (Real-time)	Simple

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Grant Lifecycle Management	Expense	VerifyExpense	Research Admin (InfoEd)	Finance ERP (Workday Financials)	InfoEd API (OAuth2)	Workday API (WS-Security)	API-led (Real-time)	Medium

5.2 Research Outputs & Publishing

Description: Researchers publish findings. Publications tracked in institutional repository and linked to grants.

Actors: Researcher, Library, Publisher

Systems: Research Admin, Institutional Repository, Publisher Portal

Sequence Diagram

```
sequenceDiagram
    participant RES as Researcher
    participant RAD as Research Admin<br/>(InfoEd)
    participant REPO as Institutional Repository<br/>(DSpace)
    participant PUB as Publisher
    RES->>PUB: Submit manuscript<br/>title=Deep Learning in Education
    PUB-->>RES: Accepted<br/>DOI: 10.1234/edu.2026.005
    RES->>RAD: Register publication<br/>DOI, journal, authors
    RAD->>REPO: Deposit full-text<br/>open access, CC-BY license
    REPO->>REPO: Assign handle<br/>handle: 1234/56789
    RAD->>RAD: Update grant output<br/>GR-2026-045, publication count+1
    RES->>REPO: View impact metrics<br/>citations=12, downloads=450
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Research Outputs & Publishing	Publication	RegisterPublication	Research Admin (InfoEd)	Institutional Repository (DSpace)	InfoEd API (OAuth2)	DSpace REST API (API Key)	API-led (Real-time)	Simple
Research Outputs & Publishing	Grant Output	UpdateGrantOutput	Research Admin (InfoEd)	Research Admin (InfoEd)	Internal	Internal	API-led (Real-time)	Simple

Institutional Engagement

Marketing, recruitment, alumni relations, fundraising, and community partnerships.

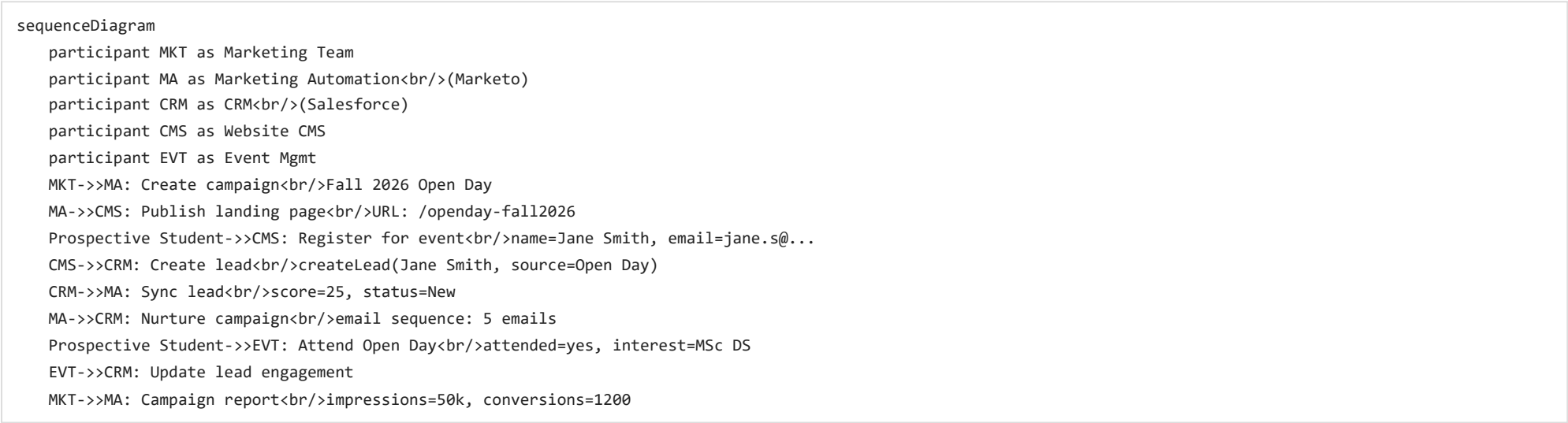
6.1 Marketing & Student Recruitment

Description: Marketing campaigns generate prospects. Inquiries route to CRM for nurturing. Events tracked. Applications originate from recruitment activities.

Actors: Marketing Team, Prospective Student, Admissions

Systems: Marketing Automation, CRM, CMS, Event Mgmt

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Marketing & Recruitment	Lead	CreateLead	Website CMS	CRM (Salesforce Education Cloud)	Webhook / REST API	Salesforce REST API (OAuth2)	Event-driven	Simple
Marketing & Recruitment	Campaign Lead	SyncCampaignLead	CRM (Salesforce Education Cloud)	Marketing Automation (Marketo)	Salesforce REST API (OAuth2)	Marketo REST API (OAuth2)	Batch (Scheduled)	Medium
Marketing & Recruitment	Event Engagement	UpdateEventEngagement	Event Mgmt System	CRM (Salesforce Education Cloud)	Event API (REST)	Salesforce REST API (OAuth2)	API-led (Real-time)	Simple

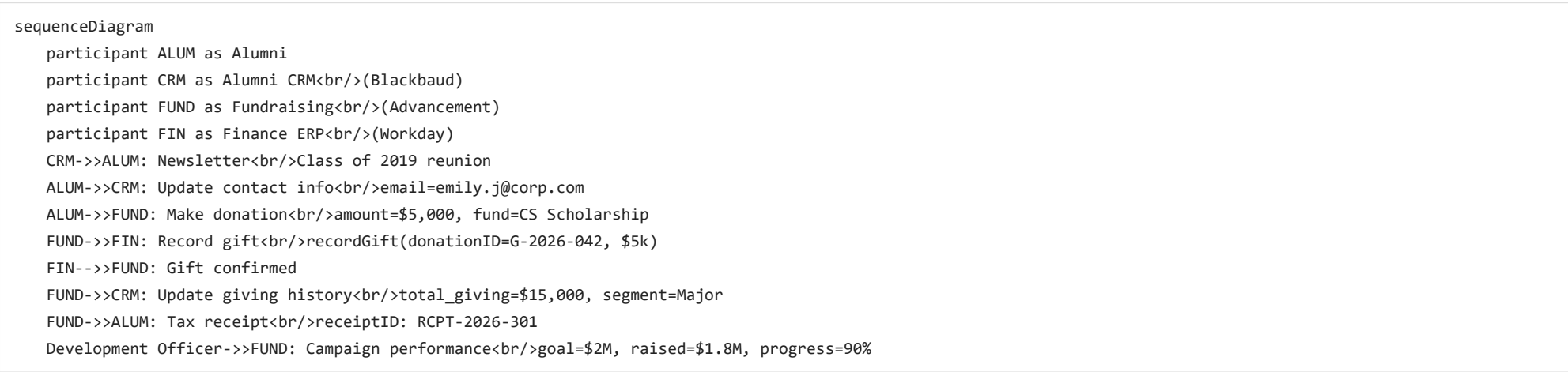
6.2 Alumni Relations & Fundraising

Description: Alumni records maintained. Fundraising campaigns target donors. Donations processed and receipts issued.

Actors: Alumni, Development Officer, Finance

Systems: Alumni CRM, Fundraising Platform, Finance ERP

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Alumni Relations & Fundraising	Donation	RecordGift	Fundraising Platform (Advancement)	Finance ERP (Workday Financials)	Advancement API (OAuth2)	Workday API (WS-Security)	API-led (Real-time)	Medium
Alumni Relations & Fundraising	Giving History	SyncGivingHistory	Fundraising Platform (Advancement)	Alumni CRM (Blackbaud)	Advancement API (OAuth2)	Blackbaud API (OAuth2)	Event-driven	Simple

Library & Learning Support

Library resource management, digital content delivery, research support, and course resource lists integrated with LMS.

7.1 Library Resource Management

Description: Library acquires, catalogs, and circulates resources. Course reading lists linked to LMS. Digital resources accessed via portal.

Actors: Librarian, Student, Faculty
Systems: Library System, LMS, Student Portal, Discovery Service

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Library Resource Management	Reading List	PushReadingList	Library System (Ex Libris Alma)	LMS (Canvas)	Alma API (API Key)	Canvas LTI / REST API (OAuth2)	API-led (Real-time)	Medium
Library Resource Management	Search Query	SearchDiscovery	Student Portal	Discovery Service (Ex Libris Primo)	Portal REST API	Primo API (API Key)	API-led (Real-time)	Simple
Library Resource Management	Usage Analytics	SyncUsageAnalytics	Library System (Ex Libris Alma)	LMS (Canvas)	Alma API (API Key)	Canvas API (OAuth2)	Batch (Scheduled)	Simple

Enterprise Data & Analytics

Centralized data integration, data warehousing, business intelligence, and institutional reporting across all domains.

8.1 Data Integration & Warehousing

Description: Data from all source systems (SIS, LMS, HR, Finance, CRM) is ingested, transformed, and loaded into the enterprise data warehouse for reporting and analytics.

Actors: Data Engineer, Data Analyst

Systems: SIS, LMS, HRIS, Finance ERP, CRM, Data Warehouse, BI Tool

Sequence Diagram

```
sequenceDiagram
    participant SIS as SIS<br/>(Ellucian Banner)
    participant LMS as LMS<br/>(Canvas)
    participant HR as HRIS<br/>(Workday HCM)
    participant FIN as Finance<br/>(Workday Financials)
    participant CRM as CRM<br/>(Salesforce)
    participant DW as Data Warehouse<br/>(Snowflake)
    participant BI as BI Tool<br/>(Tableau)
    SIS->>DW: CDC replication<br/>tables: student, course, enrollment, grade
    LMS->>DW: Daily extract<br/>tables: course_activity, assignment, quiz
    HR->>DW: Full refresh<br/>tables: employee, faculty_assignment, payroll
    FIN->>DW: Incremental load<br/>tables: invoice, payment, ledger, budget
    CRM->>DW: Hourly sync<br/>tables: lead, prospect, campaign, alumni
    DW->>DW: Transform & model<br/>star schema: fact_enrollment, dim_student
    BI->>DW: Query data<br/>DAX/MDX: retention, graduation rate
    DW-->>BI: Results
    BI->>BI: Dashboard: Institutional KPIs
    BI->>BI: Report: Student Success Analytics
```

Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Data Integration & Warehousing	Student Data	ReplicateStudent	SIS (Ellucian Banner)	Data Warehouse (Snowflake)	Banner CDC (HVR/Attunity)	Snowflake JDBC / Snowpipe	Database Replication (CDC)	High
Data Integration & Warehousing	Course Activity	ExtractCourseActivity	LMS (Canvas)	Data Warehouse (Snowflake)	Canvas API (OAuth2)	Snowflake JDBC / Snowpipe	Batch (ETL)	Medium
Data Integration & Warehousing	Employee Data	SyncEmployee	HRIS (Workday HCM)	Data Warehouse (Snowflake)	Workday API (WS-Security)	Snowflake JDBC / Snowpipe	Batch (ETL)	Medium

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Data Integration & Warehousing	Financial Data	LoadFinancialData	Finance ERP (Workday Financials)	Data Warehouse (Snowflake)	Workday API (WS-Security)	Snowflake JDBC / Snowpipe	Batch (ETL)	Medium
Data Integration & Warehousing	CRM Data	SyncCRMDData	CRM (Salesforce Education Cloud)	Data Warehouse (Snowflake)	Salesforce REST API (OAuth2)	Snowflake JDBC / Snowpipe	Batch (ETL)	Medium

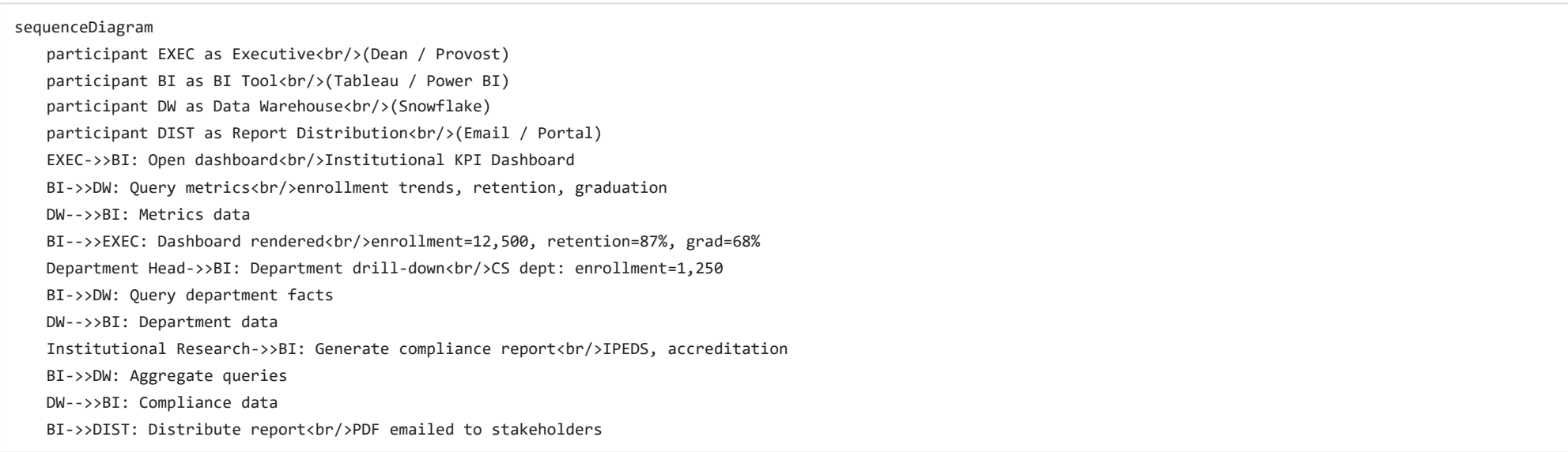
8.2 Institutional Reporting & BI

Description: Dashboards and reports for institutional KPIs: enrollment trends, retention rates, graduation rates, financial health, faculty metrics.

Actors: Executive, Department Head, Institutional Research

Systems: Data Warehouse, BI Tool, Report Distribution

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Institutional Reporting & BI	Dashboard Metrics	QueryDashboard	BI Tool (Tableau / Power BI)	Data Warehouse (Snowflake)	Tableau Server / Power BI Gateway	Snowflake ODBC / JDBC	API-led (Real-time)	Simple

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Institutional Reporting & BI	Compliance Report	GenerateComplianceReport	BI Tool (Tableau / Power BI)	Report Distribution (Email / Portal)	Tableau REST API / Power BI API	SMTP / Portal API	Batch (Scheduled)	Medium
